The service node — an advanced IN services element

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The service node is an advanced IN services element designed to allow the rapid introduction of voice and data services into the network. This paper discusses the architectural and technical issues identified during the development of a service node trial platform to support a range of PSTN services. The adopted solutions and technical direction have provided experience to support and specify the introduction of IN capability into the core network.

1. Introduction

With the accelerated trend towards more open and competitive global telecommunications networks, many traditional network operators are rapidly losing their share of the lucrative mass telephony market to new entrants. As a result of technological advances and regulatory controls, users now have a choice of communications products and services available to them.

In order to retain large users and win back customers from new entrants, network operators are being forced to provide greater flexibility in configuring their networks to make best use of available resources and provide the rapid deployment of new and innovative services.

The challenge for the major network operators is to overcome the inherent architectural limitations of a switch-based legacy network. At present, national service deployment in a switch-based network requires all switching manufacturers to develop the same features in a transparent way. Due to the logistical and commercial issues associated with such an upgrade, the process is costly and time consuming.

As an alternative, many network operators have invested in overlay networks to provide flexible new services to leading edge customers. While improving responsiveness to market forces in the short term, overlay networks result in fragmented access and management of the overall network.

Due to these difficulties, network operators have opted for the long-term aim of developing a core intelligent network (IN) capability. The concept of the IN provides major advances in the provision of services to the user while allowing the service provider increased ease and flexibility in deployment. The IN standards and architectural issues are discussed elsewhere [1].

Having identified the IN as the target infrastructure that will provide the competitive edge in an advanced telecommunications market, the network operator must address the non-trivial task of migrating services from an embedded switch-based PSTN to a core IN architecture. The key changes that must be made include:

- increasing the signalling functionality of the PSTN to handle the additional traffic associated with IN-based services,
- adopting an open systems interface to support service-independent signalling protocols between the various IN elements.

The latter must conform to international standards and is of critical importance if IN-based services are to operate transparently across equipment provided by multiple vendors.

Successful migration from the existing switch-based services to a distributed IN architecture will only be possible following a detailed analysis of the various network constraints. These include minimising service development costs, ensuring the least possible impact on existing PSTN resources, and allowing the smooth introduction of services with gradual traffic growth. In addition, possible evolution in services and the likely emergence of unforeseen demand for some services must be considered.

Working in collaboration with researchers at BT Laboratories, who have been involved in shaping the future architecture of the BT core IN and the evolving international standards within CCITT and ETSI, the network intelligence engineering centre embarked on a development programme to produce a self-contained IN element known as a service node. The main objective of the service node (SN) development was to deliver, within extremely short timescales, a network-worthy platform to trial certain IN-based services within the PSTN. Recognising that the core IN will take some years to realise, investment was made in the SN to gain early insight into the technical and architectural issues associated with the implementation of IN.

Based on an open system architecture, an SN is ideally suited for assessing the feasibility of many IN concepts and services by rapid integration of emerging best-of-breed technologies. By performing trials of certain IN-based services on a selected customer population within a live PSTN environment, this study provided BT with the ability to prototype and field trial IN-based services and concepts in advance of the core IN development.

This paper discusses some of the technical and architectural issues identified during the development of an SN into a network-worthy advanced IN-based services element. The deployment of the SN to trial certain IN-based services is also discussed. The paper concludes by exploring the possible application and reuse of SN technology within BT's core IN.

2. Key requirements

Some of the key requirements adopted during the design of an SN were based on the following prerequisite characteristics of an IN element:

- extensive use of information processing,
- efficient use of network resources,
- modular, reusable and scalable,
- conforms to standardised, service-independent protocols and communications,
- multivendor capability,
- multinetwork capability,
- rapid service delivery,
- service deployment.

This SN design philosophy has resulted in a generic framework for supporting advanced services. By adopting existing, established computer technology to provide the underlying control and processing capability, an SN architecture readily supports an open interface which allows for the ability to procure and integrate commercially available, third party network resources. This approach is consistent with the IN objective for rapid service creation and minimal re-engineering of network elements to support new services.

A major requirement is that an SN should support a generic service-independent framework, whereby new resources (e.g. interactive voice response, voice recognition, facsimile store and forward, screen phone, audio-conferencing, e-mail) can be easily integrated to provide additional service capabilities.

The latter implies an inherent flexibility to support an easily scalable and modular design. This functionality would enable BT to meet the specific needs of its corporate customers by providing specialised services, possibly with a short life expectancy, in support of a customer's business strategy.

In the likely emergence of unforeseen demand for some trial services, an SN should support a strategy which allows the timely roll-out of the service post-trial, while extending the customer base of the service. Alternative scenarios can be depicted where an SN is deployed to support extended marketing trials or to provide BT with the ability to support a selective marketing strategy during a transition to the core IN environment.

Many other requirements that impacted the design of the SN architecture included performance engineering issues, continuous availability, high throughput, real time call-handling, dynamic reconfiguration, platform scalability, service independence, and the ability to support multiple network interfaces.

3. Functional prototyping and system specification

This section summarises the results of the rapid prototyping phase of the SN development. This exercise was used to validate, verify and refine SN requirements. The main objectives included:

- verification of the SN functional requirements,
- a detailed definition of the SN software and hardware requirements,
- development and evaluation of a functional SN prototype to determine quantitative and qualitative benchmark and design criteria.

Based on previous studies [2], the system architecture adopted a distributed multiprocessor architecture based around the UNIX operating system. For improved performance, each of the major subsystems had to be designed to operate efficiently in a multitasking and, where available, multiprocessing environment.

A key requirement of the control processor, which forms the core processing capability of an SN, was the ability to prioritise and schedule application/network requests in real time. This capability was essential if a predetermined quality and responsiveness of service was to be offered to customers.

Significant performance benefits could be attained by writing customised memory management functions to override the generic functionality offered by the operating system. By optimising memory performance, the control processor could minimise the level of context switching between SN processes, thereby increasing overall efficiency and throughput.

The decision to adopt UNIX as the underlying operating system is based on the results from previous evaluation studies [2, 3]. The key benefits of UNIX, within a multi-service telephony environment, include widespread multi-vendor support, ease of portability of applications and the flexibility of having supplier-independent target platforms.

However, a number of limitations were identified in the UNIX kernel during the performance benchmarking of the SN prototype. The main problem was the single-threaded design of the UNIX scheduler, which adversely affected initial estimates of system performance. Further tests on a range of UNIX-based products showed this to be a common limitation to all of the systems benchmarked.

Despite these problems, the UNIX kernel is more than adequate for the level of processing required to achieve current market predictions for services. However, these problems have been notified to the relevant suppliers and this bottleneck should be eliminated in future releases of UNIX.

4. Architecture overview

At the core of an SN architecture is a network database much like a service control point (SCP), but it also contains a transport interface and peripherals, similar to an intelligent peripheral (IP), which can be used to facilitate user interactions with the network. Figure 1 illustrates the SN architecture.

A brief summary of each of the hardware components is given below:

- control processor — comprises a fault-tolerant commercially available computer system running the UNIX operating system,

- local area network (LAN) (multiple ethernet) — a typical system configuration requires a minimum of two ethernets for security, and can currently support a maximum of eight,

- resource/switch interface — European primary rate multiplex hierarchy standard E1 (USA standard T1 also in development),

- system console — a maximum of four consoles, which can be located locally or remotely, can currently be supported,

- CCITT signalling No. 7 (SS7) — the current SS7 implementation is closely coupled with the control processor hardware,

- switch — the switch currently supports 256 30-channel PCM systems, multi-frequency detection being supported on every channel (a number of exchange tones are also provided by the switch as a generic capability),

- resources — resources are provided using an N+1 sparing policy.

The SN switch is a core component that serves as a concentrator to provide access to a family of resources. The resources can provide such voice-service functions as dual tone multifrequency (DTMF) detection, speech synthesis, speech recording and playback, voice recognition and a broad range of other facilities.

Figure 2 shows the major software subsystems of the SN.

The key components of the control processor will be described in later sections.

5. Continuous availability

A key customer requirement is that an SN should provide continuous availability. This characteristic has a major impact on the overall cost of the platform. Consequently, very careful consideration was given to the level of resilience and fault tolerance required of the various processing elements that make up an SN. Certain criteria were devised to compare the mission-critical components against economic factors and performance.

The mission-critical components within an SN were identified to be:

- control processor — controls the service logic and platform resources,

- network interface — handles the lower level SS7 signalling,

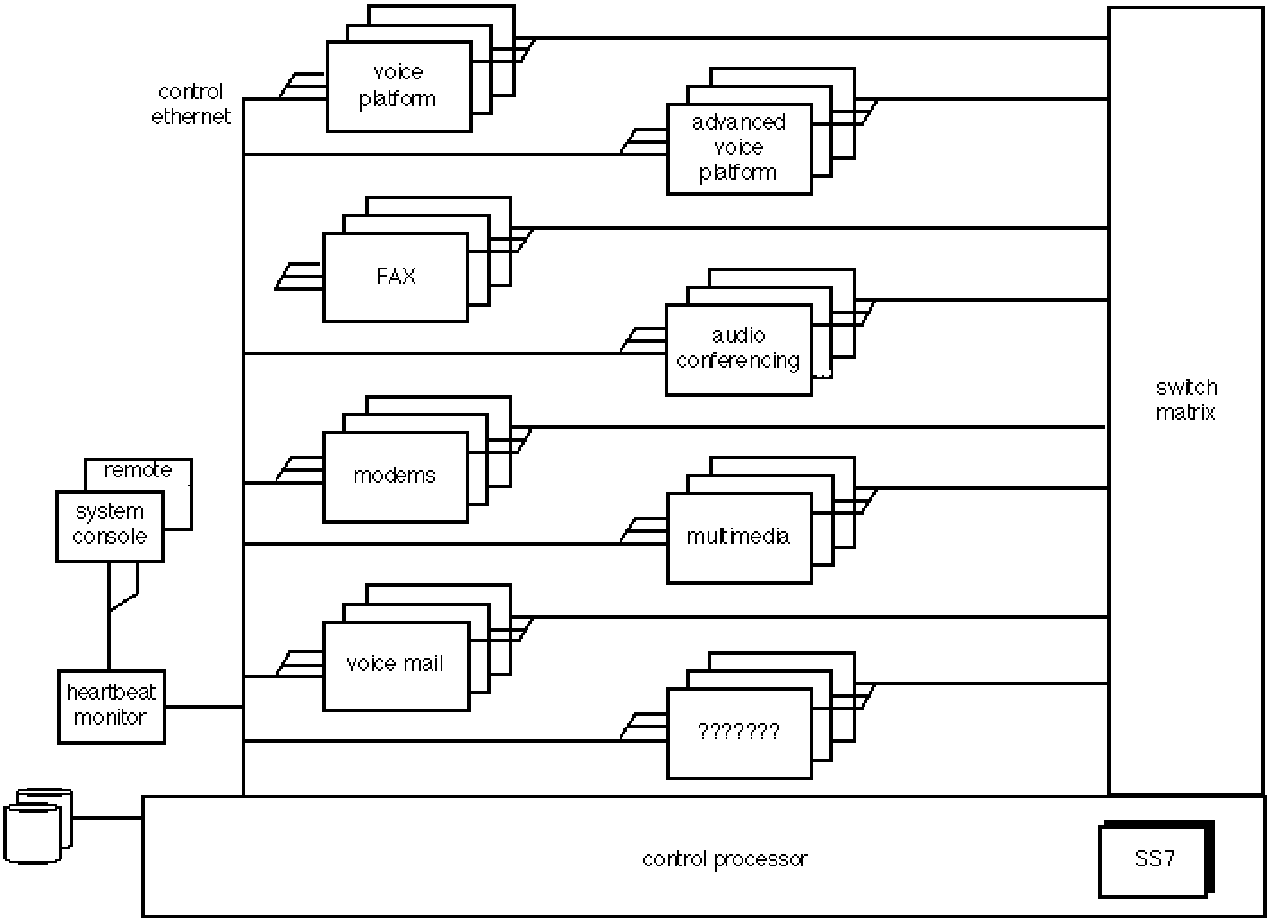

Fig 1 The SN architecture.

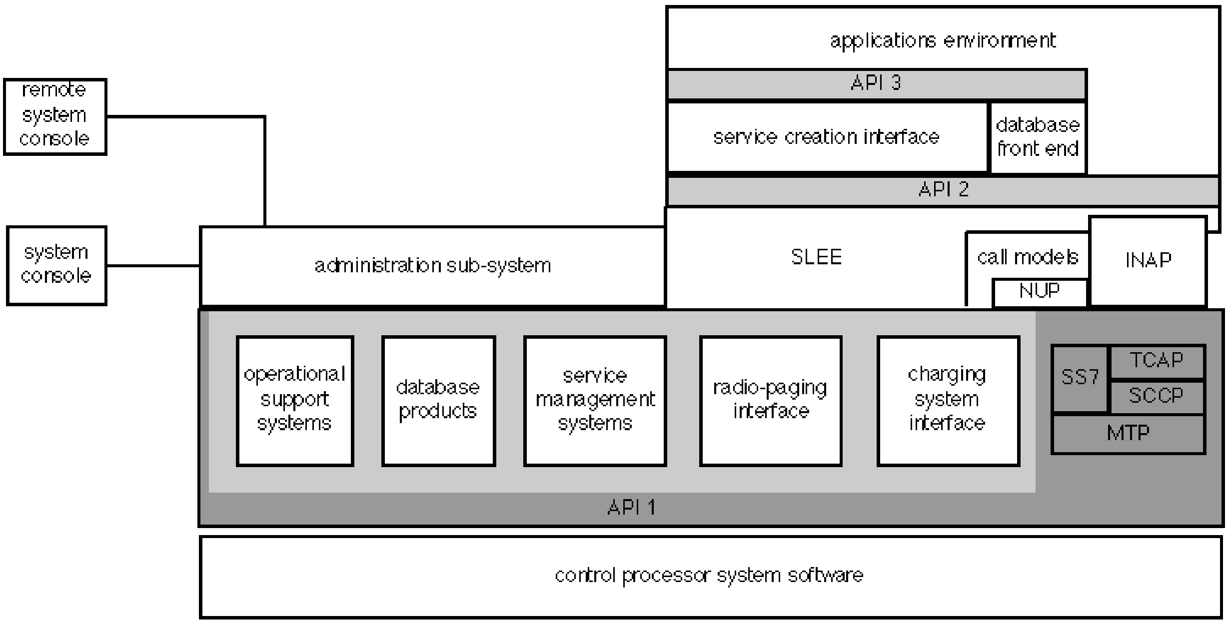

Fig 2 Control processor overview.

- switch matrix — handles the routeing of signalling and speech into the platform,

- inter-process communications — enables the reliable transport of messages between the control processor, switch and SN resources.

The required system behaviour of these components leads to a system architecture that must embody hardware reliability and flexibility. The SN provides continuous availability by incorporating a combination of (high-cost) hardware fault tolerance and (low-cost) N+1 redundancy techniques.

The SN control processor (and integrated network interface) is based upon a commercially available, fault-tolerant hardware architecture and operating system. Despite the high capital cost, this approach greatly simplified the design effort required to implement the sub-systems that were resident on the control processor. The designers of the SS7, service logic execution environment (SLEE), SN accounting and charging (SNAC), and node manager subsystems, could rely on the underlying hardware environment to sustain system availability.

The SN switch matrix adopts a combination of hardware fault tolerance and N+1 sparing to provide high availability.

A critical element of an SN architecture is the internal LAN which carries the inter-process communications between the various subsystems. This mechanism was implemented using a dual-ethernet configuration to provide the necessary resilience. In addition, a new software protocol was developed above the UDP/IP stack to provide assured LAN reliability, performance and responsiveness.

The SN architecture limits the effect of a single point of failure such that it does not propagate throughout the system to adversely degrade performance. This characteristic is further enhanced by designing a secure resource allocation and control mechanism within the SLEE. Consequently, a given resource can only be accessed through a well-defined interface, which guarantees that any such access will not corrupt nor cause failure in any other SN component.

6. Human factors implications

The service node architecture and functionality is defined by a set of service-independent functional building blocks (FBB) distributed across the SN and its resources. The scope of each FBB was designed to be fully configurable while limiting the capability of each block. Typical FBBs include detect incoming call, receive digits, store digits, analyse digits, route call, check call status, play announcement, perform simple speech recognition and free speech circuits. To accommodate a wide variety of services of different complexity, the FBB concept provides highly flexible and reusable software components with specific service-independent capabilities. Each FBB has set input parameters that specify the required functionality, e.g. these could include the text to be played to the user during a voice prompt, or to specify the structure of any information required as input from the calling customer.

The rationale of the FBB was to reuse existing building-blocks on the SN resources, thereby reducing software development and maintenance costs. This approach is fully consistent with the IN objective of rapid service deployment.

The main service that was identified for benchmarking the SN is known as personal numbering (PN). This service provides the customer with the ability to register a series of physical telephone numbers to a single, logical personal number. Each customer has a personal profile maintained within the SN database which can be updated on a daily basis to register the desired time-of-day routeing. The customer is also given a default number to which calls can be routed should they need to deviate from their planned itinerary. An integrated voice-mail system and a number of ancillary support services were also provided to assist with customer mobility.

The PN service provides a sophisticated call completion facility, whereby the calling party can hunt up to three customer-configurable telephone numbers (on busy or no-answer) before calls are terminated to voice-messaging.

The interface provided to the PN customer when accessing the service to either configure their profile, or to access the personal number service feature set, took the following form:

- initially the customer is given a greeting and requested to enter their account number and PIN for authorisation (this is designed to provide security and fraud limitation to BT and its customers),

- if the customer is successful, the SN will play the following announcement: “Good <time of day>. Welcome to the BT intelligent peripheral demonstration. You have <number> new voice message(s)”,

- the customer would then hear the main menu for the service.

Using the FBBs, this entire prompt could be requested using a single FBB request to the speech resource for each speech segment within the prompt.

7. The network interface

The SN control processor contains a set of integral fault-tolerant network interface cards which are used to perform portions of the SS7 signalling protocol — in particular, the message transfer part (MTP), signalling connection control part (SCCP), and transaction capabilities application part (TCAP) layers.

The SS7 capability of the control processor was of major importance in the selected implementation strategy. In addition, the SS7 had to support open and stable interfaces. Using the defined interfaces, customised user parts could be developed that could be transparently integrated into the SN environment.

A key development that had to be undertaken was BT's national user part (NUP), which handles PSTN call set-up and call processing. The NUP provides direct circuit-related signalling and control. Ongoing developments include the implementation of ETSI capability set 1 (CS-1) IN application part (INAP) and the mobile application part (MAP) protocols. CS-1 INAP will provide the SN with the capability to perform connectionless IN-related functionality. The MAP will provide the ability to interconnect with the cellular networks, so that the SN can provide transparent multinetwork access.

The SN has the capability to support multiple application user parts to allow easy interconnect to a host of other licensed operator (OLO) networks.

Basic link management and configuration functions have been integrated into the SN system console via the node manager subsystem. The approach taken for the integration of the SS7 subsystem is typical of the approach adopted for the integration of third party products throughout the development programme.

8. The resource interface

The SN provides a consistent and secure method of integrating new resources into the node. By adopting a layered SLEE, the service logic was isolated from the SN resources and lower-level basic call processing. By abstracting the service logic from the underlying platform architecture, new service features can be easily added without having to upgrade any of the core SN software subsystems.

The SLEE can be regarded as a real time call processing system, autonomously handling basic telephony and providing an open interface to additional features for supplementary services. The application is a purely sequential program that has a completely synchronous, transaction-oriented interface with the SLEE. In this way, self-contained applications can be readily developed, the functionality of which is via system calls to the SLEE.

The SLEE keeps track of and detects events and conditions that require an application instance. An application is usually triggered either by an event or a condition at a predetermined point in a call. Once a trigger is detected, the SLEE will schedule the application to take control of a call.

The remainder of this section will present an example of service requiring a customer interaction with a speech

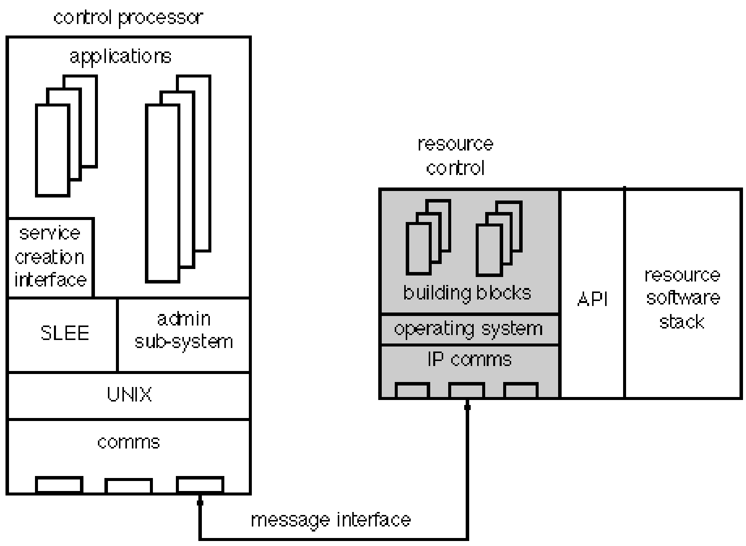

Fig 2 Control processor resource interface.

resource. The example will be limited to the message flows over the interface shown in Fig 3.

- The application makes a SLEE API_Reserve_Resource request to play an announcement and receive DTMF digits. The application is allowed to specify the name it will use to reference the resource. Thus, the application programmer can adopt their preferred resource naming convention within a specific service.

- The SLEE will then issue an API_Connect to the switch requesting that the incoming call be connected to the specified resource.

- The selected resource must also be instructed to perform the required function. In this example, the resource must play an announcement and receive digits. The SLEE constructs the API command, using the appropriate parameters, and despatches the request to the resource.

- Once the caller is connected to the resource, the SLEE will issue a Start_Dialogue to the resource which will then play the selected announcement.

The SLEE philosophy provides application developers with a platform-independent method of constructing services, minimising the development effort required to construct a service.

A further benefit of the SLEE is that it minimises any degradation in performance due to inadvertent faults in SN resources. For example, if a particular resource is causing a processing bottleneck, the SLEE resource allocation algorithm can be used to smooth over such difficulties to deliver an acceptable level of responsiveness to the customer.

9. Flexible accounting and charging

A key objective of IN is the capability to allow the service provider to maintain control of routeing decisions for services, based upon time of day, day of week, authorisation codes, etc. The routeing decision criteria will be under the control of the service provider. Similarly, charging decisions can be based on locations, destinations, time of day, authorisation codes, etc. The service provider will require similar flexibility to define the desired charging criteria for every aspect of a service.

The SN accounting and charging (SNAC) system is the centralised service node system that produces platform-independent usage records (PIUR) containing customer service usage data. The application developer is provided with a flexible mechanism for generating:

- charge records for customer billing and invoicing,

- statistics data for marketing analysis and platform auditing.

The SNAC data generation function (distributed across the application, SLEE, and call model) produces raw customer service usage data when specified generation criteria have been fulfilled. Once a complete service-independent usage record (SIUR) has been built, the SLEE will forward the SIUR to the SNAC for storage, formatting, and outputting to the relevant user.

The SNAC provides the platform with the capability to charge for the following generic charge events:

- incoming call events — based on the duration of a call to a specific platform feature or resource,

- outgoing call events — based on the duration of an external call to a specific destination or platform,

- elapsed time events — based on the duration of a call whilst using a specific platform feature or resource,

- one-off events — based on the selection of a specific platform feature or resource, for example, activating a radiopager.

Many other parameters, including absolute timings, geographic routeing, type of resource used, etc, will also be collected by the platform and made available to the billing system, such that the tariffing and costing algorithm used to calculate the actual cost can take into consideration all data associated with a call.

In order to support IN-based services, the SNAC can cope with a wide range of charging and accounting data depending on the type of resources accessible to a service. It is not sufficient to simply generate new call-record structures to support new services.

Specifically, the SNAC accounting capability can generate:

- customer and service profiles, which can be used by a marketing service for assessing the efficiency and effectiveness of the service or platform,

- detailed customer service usage that occurs during a single instance of a service, resulting in a complete account of usage,

- aggregate customer service usage across multiple instances of service that have been provided within a specified aggregation interval,

- usage data formatted with full or partial call-records and output to the destination billing system.

The SNAC has the level of flexibility to support new services and any associated call-record formats on an evolutionary basis.

10. Interconnect

The cost-effective realisation and timely delivery of services are essential requirements for network-deployed IN platforms. Hence, the IN platform must be adaptable to ensure interconnection and communication with a broad range of existing network elements. The SN adopts a flex-ible interfacing technique between key sub-systems such as the SLEE, SS7 NUP, SNAC, and special-ised on-line customer databases. This allows the SN to rapidly 'graft' interfaces to new platform resources in a very short time interval.

To achieve this, a generic conversational resource interface was designed into the SLEE. This enables the easy introduction of new interfaces with minimal disruption. This interface is extremely flexible and currently supports interfaces to a radiopaging centre, service management console, and cashless call validation/billing database. More recently, development has commenced on the SS7 INAP and MAP protocols taking full advantage of this interface.

11. Discussion and conclusions

This paper has reviewed the issues associated with the design, development, and deployment of a service node. The main objective of this project was to deliver an advanced IN services element for trialling certain IN-based services and concepts within a live network.

Many issues have been identified during the development of the SN and the subsequent trial of selected services. Having demonstrated that an SN architecture can fulfil the main objectives of IN, the service trials have provided valuable insight into the issues associated with deploying IN.

The fundamental IN architecture depicts a fully distributed core network comprising SSPs, SCPs and IPs. The interconnect and messaging between these entities is via SS7 signalling links. It is envisaged that, for a typical IN-based service, such as a cashless services call, there will be in the order of ten or more complex INAP operations per transaction. Assuming an average SCP capacity of 500 transactions per second, during periods of heavy loading, this architecture will result in a significant amount of SCP-SSP-IP messaging. The likely observable outcome is an added signalling delay and customer perceived degradation in service. Overall performance and call throughput is further compromised by the additional processing overhead due to formatting, timing of responses, and verification of the received information.

The SN architecture overcomes this potential bottleneck by tightly coupling the SCP and IP functional entities into a single physical network element. Although the SCP-IP protocol remains conformant to the CCITT standards, the transport layer is based on high-speed UNIX inter-process communications. Using the accounting capability of the SN, future studies can be used to investigate the performance engineering issues associated with this messaging interface.

A major area of concern for many network operators is the potential loss of revenue due to fraudulent users. Many operators now consider fraud detection as a major system requirement. Preliminary studies have already been performed to enhance the existing SN security features to provide fraud detection. Changes to the method of SNAC charge record generation and an integrated intelligent trend analysis subsystem could form the basis of a fully functional test-bed for fraud detection and analysis.

In its current state, the service node fulfils many of the objectives of the IN concept. A service node encapsulates the level of flexibility and adaptability required to minimise the lead time to market of new and innovative services. This study has provided the ability to prototype and field trial IN-based services and concepts in advance of the deployment of core network IN facilities. Services have been trialled at low risk using low traffic volumes which, where successful, can be migrated to the core IN for national roll-out.

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